```
clc;
clear;
close all;

% FIR Filter Parameters
b = [1 4 6 4 1]/16;
Fs = 1000;
N = 128;

x = [1 zeros(1,N-1)];
n = 0:N-1;

% Direct Realization
yd = filter(b,1,x);

% Cascade Realization
s1 = [1 2 1];
s2 = [1 2 1];

yc = filter(s1,1,x);
yc = filter(s2,1,yc);
yc = yc/16;

% Symmetry and Group Delay
symErr = max(abs(b-fliplr(b)));
D = (length(b)-1)/2;

if symErr < 1e-12
    symmetry = 'Symmetric';
else
    symmetry = 'Not Symmetric';
end

% Frequency Response
[H,w] = freqz(b,1,1024);
phase = unwrap(angle(H));
gd = grpdelay(b,1,1024);

% Numerical Comparison
err = max(abs(yd-yc));
dev = err/max(abs(yd))*100;

% Display Results
fprintf('\nFIR FILTER REALIZATION RESULTS\n');
fprintf('Sampling Frequency = %d Hz\n',Fs);
fprintf('Number of Samples = %d\n',N);
fprintf('Coefficient Symmetry = %s\n',symmetry);
fprintf('Symmetry Error = %.3e\n',symErr);
fprintf('Group Delay = %.2f samples\n',D);
fprintf('Maximum Error = %.3e\n',err);
fprintf('Percentage Deviation = %.3e percent\n',dev);
```

```
% Display First 10 Samples
fprintf('\nFirst 10 Impulse Response Values\n');
fprintf(' n       Direct       Cascade\n');

for k = 1:10
    fprintf('%2d    %10.6f    %10.6f\n',n(k),yd(k),yc(k));
end

% Output Table
Parameter = {
    'Direct Realization'
    'Cascade Realization'
    'Coefficient Symmetry'
    'Group Delay'
    'Maximum Error'
};

Value = {
    'FIR filtering'
    'Cascade filtering'
    symmetry
    sprintf('%.2f samples',D)
    sprintf('%.3e',err)
};

OutputTable = table(Parameter,Value);
disp(OutputTable);

% Graph 1: Impulse Response
figure;
stem(n,yd,'filled');
hold on;
stem(n,yc,'r--');
grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Direct and Cascade FIR Impulse Responses');
legend('Direct','Cascade');

% Graph 2: Magnitude Response
figure;
plot(w/pi,abs(H),'LineWidth',1.5);
grid on;
xlabel('Normalized Frequency');
ylabel('Magnitude');
title('FIR Magnitude Response');

% Graph 3: Phase Response
figure;
plot(w/pi,phase,'LineWidth',1.5);
grid on;
xlabel('Normalized Frequency');
ylabel('Phase in Radians');
title('FIR Phase Response');
```

```
% Graph 4: Group Delay
figure;
plot(w/pi,gd,'LineWidth',1.5);
grid on;
xlabel('Normalized Frequency');
ylabel('Group Delay in Samples');
title('FIR Group Delay');

% Graph 5: Numerical Error
figure;
plot(n,abs(yd-yc),'LineWidth',1.5);
grid on;
xlabel('Sample n');
ylabel('Absolute Error');
title('Direct vs Cascade Numerical Error');

% Final Result
fprintf('\nFINAL RESULT\n');
fprintf('Direct FIR realization completed.\n');
fprintf('Cascade FIR realization completed.\n');
fprintf('Coefficients are %s.\n',symmetry);
fprintf('Group delay is %.2f samples.\n',D);
fprintf('Maximum error is %.3e.\n',err);
fprintf('Both realizations are equivalent.\n');
```

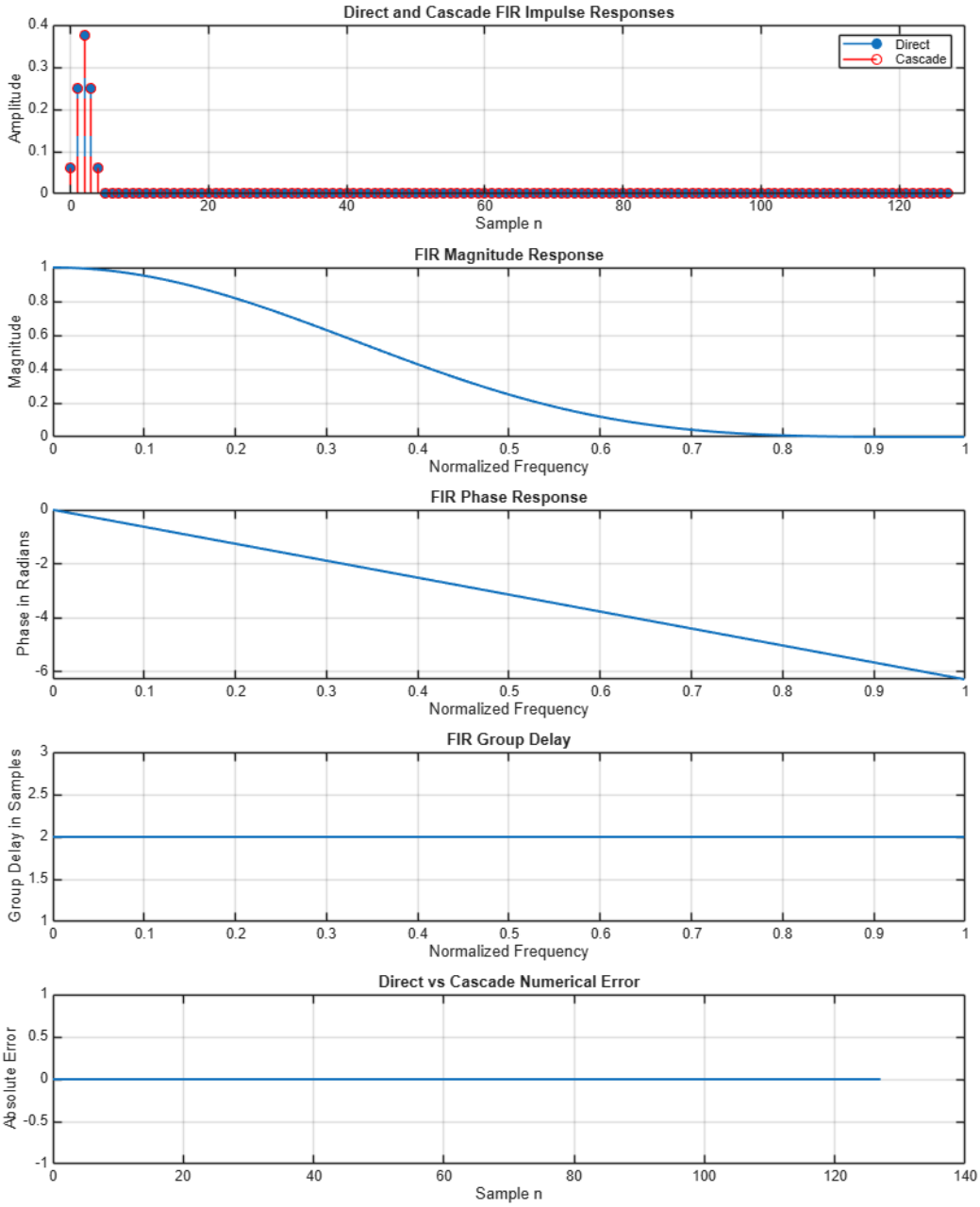
```
FIR FILTER REALIZATION RESULTS
Sampling Frequency = 1000 Hz
Number of Samples = 128
Coefficient Symmetry = Symmetric
Symmetry Error = 0.000e+00
Group Delay = 2.00 samples
Maximum Error = 0.000e+00
Percentage Deviation = 0.000e+00 percent

First 10 Impulse Response Values
 n       Direct       Cascade
 0      0.062500      0.062500
 1      0.250000      0.250000
 2      0.375000      0.375000
 3      0.250000      0.250000
 4      0.062500      0.062500
 5      0.000000      0.000000
 6      0.000000      0.000000
 7      0.000000      0.000000
 8      0.000000      0.000000
 9      0.000000      0.000000
            Parameter                    Value
    ____________________________    ____________________

    {'Direct Realization'  }    {'FIR filtering'    }
    {'Cascade Realization' }    {'Cascade filtering'}
    {'Coefficient Symmetry'}    {'Symmetric'        }
```

```
    {'Group Delay'          }    {'2.00 samples'     }
    {'Maximum Error'        }    {'0.000e+00'        }
```

```
FINAL RESULT
Direct FIR realization completed.
Cascade FIR realization completed.
Coefficients are Symmetric.
Group delay is 2.00 samples.
Maximum error is 0.000e+00.
Both realizations are equivalent.
```



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